# Human-Development-Index (HDI) Predictor

## Project Description:

The **Human Development Index (HDI) Predictor** is a Machine Learning-based web application developed using **Python, Flask, Scikit-learn, Pandas, NumPy, Matplotlib, and Seaborn**. The application predicts the Human Development Index (HDI) score of a country based on important socio-economic indicators such as **Life Expectancy**, **Mean Years of Schooling**, **Expected Years of Schooling**, and **Gross National Income (GNI) per Capita**.

The Human Development Index is a composite statistical measure introduced by the United Nations Development Programme (UNDP) to evaluate the overall development of countries. Instead of focusing only on economic growth, HDI considers human well-being by combining health, education, and living standards into a single index.The project follows the complete Machine Learning workflow including dataset preprocessing, exploratory data analysis, visualization, model training using Linear Regression, model evaluation, serialization using Pickle, and deployment through a Flask web application. The application provides an easy-to-use interface where users can enter development indicators and instantly obtain the predicted HDI score.

## Scenarios :

### Scenario 1: Predicting Very High Human Development

A government analyst enters high values for life expectancy, education indicators, and Gross National Income. The machine learning model predicts a Very High HDI Score, indicating that the country belongs to the highest level of human development.

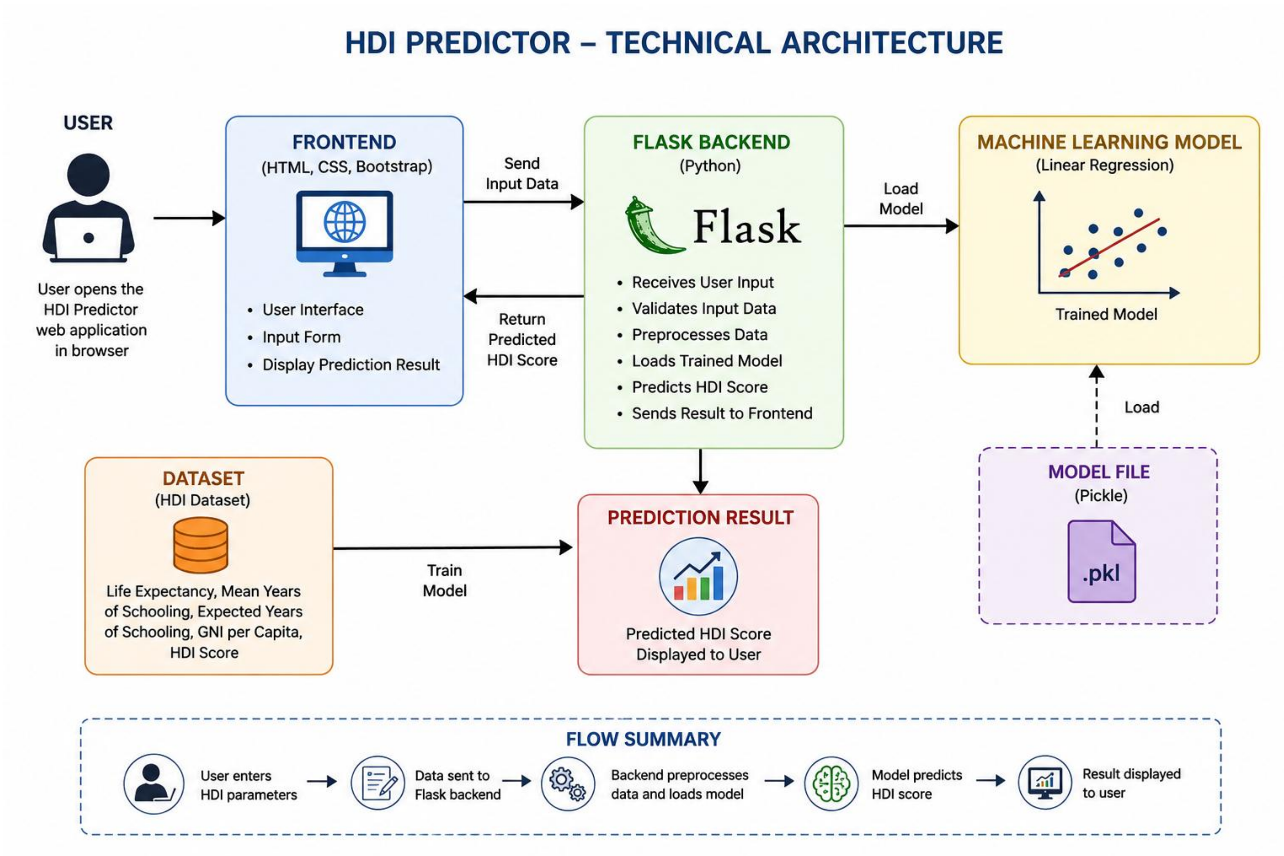
### Scenario 2: Evaluating Developing Countries

A policymaker enters moderate values for healthcare, education, and income indicators representing an emerging economy. The application predicts a **Medium HDI Score**, helping identify sectors that require improvement for sustainable development.

### Scenario 3: Identifying Low Human Development

A researcher enters lower values for life expectancy, schooling, and income. The model predicts a **Low HDI Score**, enabling organizations to recognize countries requiring policy intervention and developmental support.

## Technical Architecture:




## Architecture Description

- The frontend collects development indicator values from the user.
- Flask receives the input and validates the values.
- The trained Pickle model is loaded.
- The machine learning model predicts the HDI score.
- Flask returns the prediction to the web page.
- The predicted HDI score is displayed to the user.

## Pre-requisites

- Python Programming , Flask Framework , Machine Learning Basics
- NumPy , Pandas, Matplotlib , Seaborn , Scikit-learn , Pickle
- HTML , CSS , Bootstrap

## Milestone 1: Environment Setup & Dataset Collection

This milestone focuses on setting up the development environment and collecting the Human Development Index dataset required for model development.

### Activity 1.1: Environment Setup

- Install Python and required libraries.
- Configure Flask and machine learning packages.
- Create the project structure.

### Activity 1.2: Dataset Collection

- Download the HDI dataset.
- Study dataset features and target variable.
- Analyze the dataset for machine learning.

## Milestone 2: Data Preprocessing & Model Development

This milestone focuses on preparing the dataset and building the machine learning model for HDI prediction.

### Activity 2.1: Data Preprocessing

- Handle missing values.
- Select input and output variables.
- Split data into training and testing datasets.

### Activity 2.2: Machine Learning Model

- Train the Linear Regression model.
- Evaluate model performance.
- Save the trained model using Pickle.

## Milestone 3: Flask Application Development

This milestone focuses on developing the Flask application that connects the user interface with the trained machine learning model.

## Activity 3.1: Writing the Main Application Logic in app.py

### Define the Core Routes

- / – Displays the Home page.
- **/predict** – Accepts user input and predicts the HDI score.

### Set Up Route Handlers

- Receive user input from the prediction form.
- Validate input values.
- Load the trained Pickle model.
- Generate and display the predicted HDI score.

## Milestone 4: Frontend Development

This milestone focuses on designing an interactive and user-friendly interface for the HDI Predictor.

## Activity 4.1: Designing the User Interface

- Create responsive web pages using HTML and CSS.
- Design the input form and result page.
- Improve user experience with a clean layout.

## Activity 4.2: Display Prediction Results

- Collect user inputs.
- Send data to the Flask backend.
- Display the predicted HDI score.

## Milestone 5: Application Integration and Prediction

This milestone focuses on integrating the trained Linear Regression model with the Flask application and providing a user-friendly interface for predicting the Human Development Index.

## Activity 5.1: Interface Development

- Design the prediction page.
- Collect development indicator values.

## Activity 5.2: Prediction Processing

- Validate user inputs.
- Load the serialized model.
- Predict the HDI score.
- Display the prediction result.

## Milestone 6: Deployment & Testing

This milestone focuses on deploying and testing the application.

## Activity 6.1: Local Deployment

- Install project dependencies.
- Run the Flask application.
- Verify application functionality.

Create Virtual Environment.

```
python -m venv myenv
```

Activate Environment.

```
myenv\Scripts\activate
```

Install dependencies.

```
pip install -r requirements.txt
```

## Activity 6.2: Application Testing

- Test prediction accuracy.
- Validate user inputs.
- Ensure correct result display.

Run Flask Application:

```
python app.py
```

Open browser.

http://127.0.0.1:5000

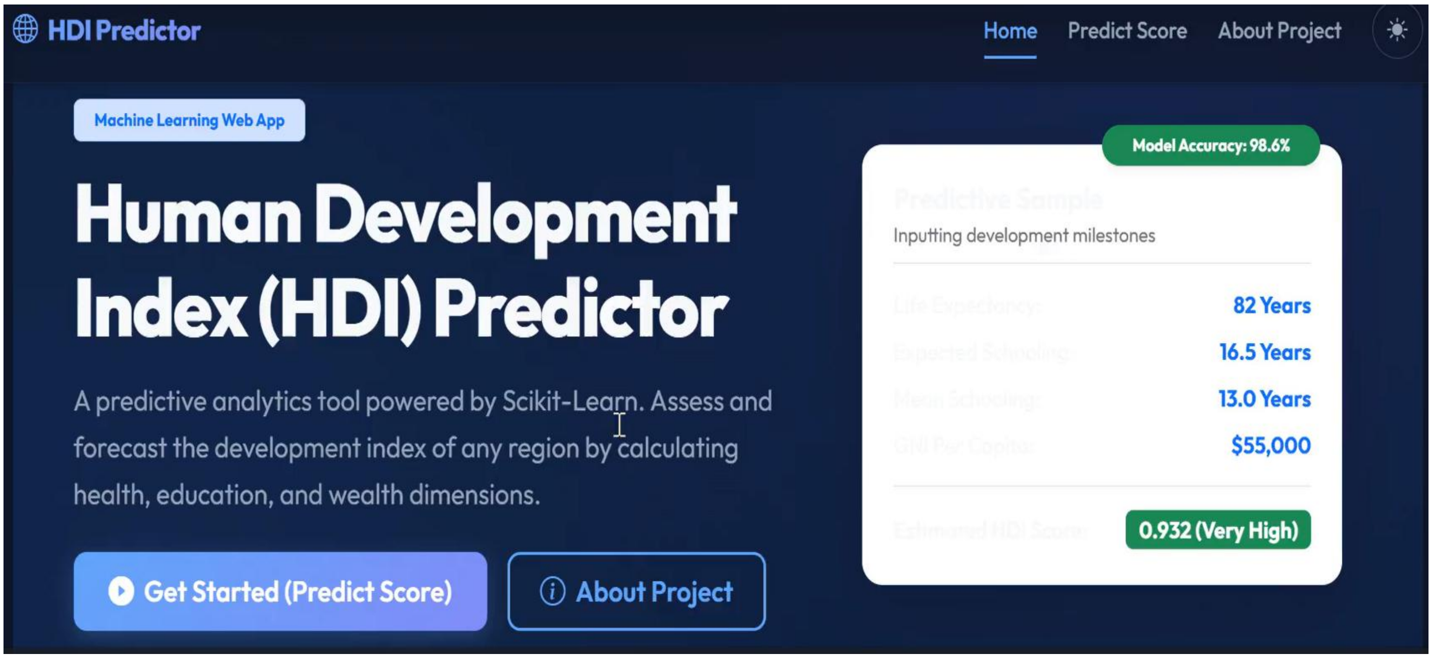
## Milestone 7: Conclusion

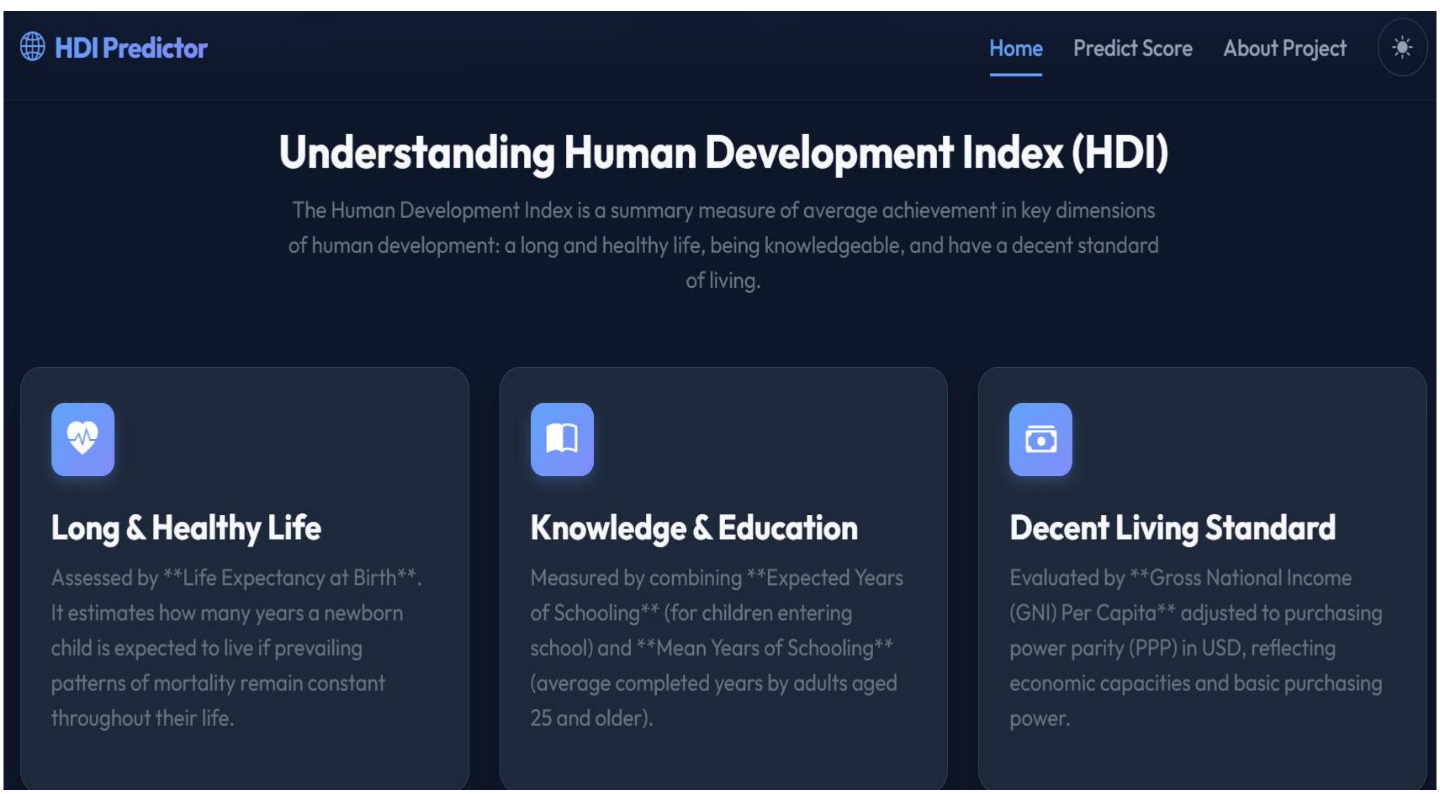
The Human Development Index (HDI) Predictor successfully integrates Machine Learning and Flask to provide accurate HDI predictions based on socio-economic indicators. The project demonstrates data preprocessing, model development, web application integration, and prediction through a simple and user-friendly interface.

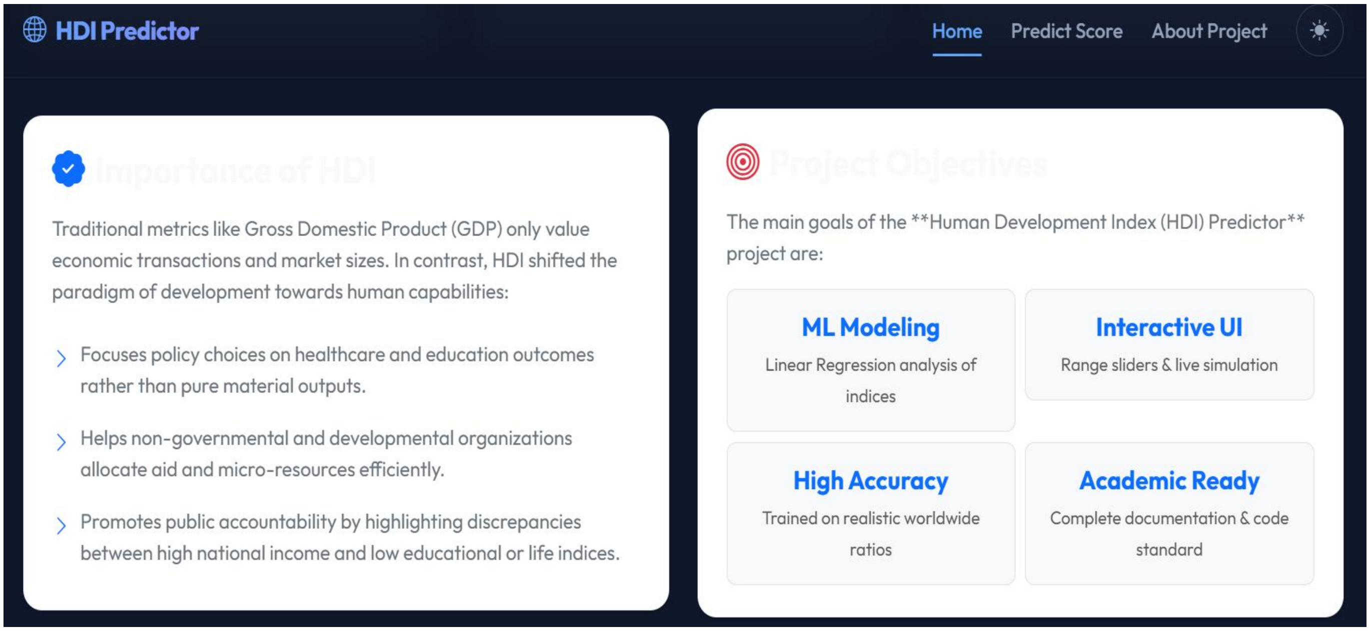
This format closely matches the style of your sample documentation and leaves room for adding screenshots under each activity.

# Exploring the Website's Web Pages:

## Home / Generator Page:



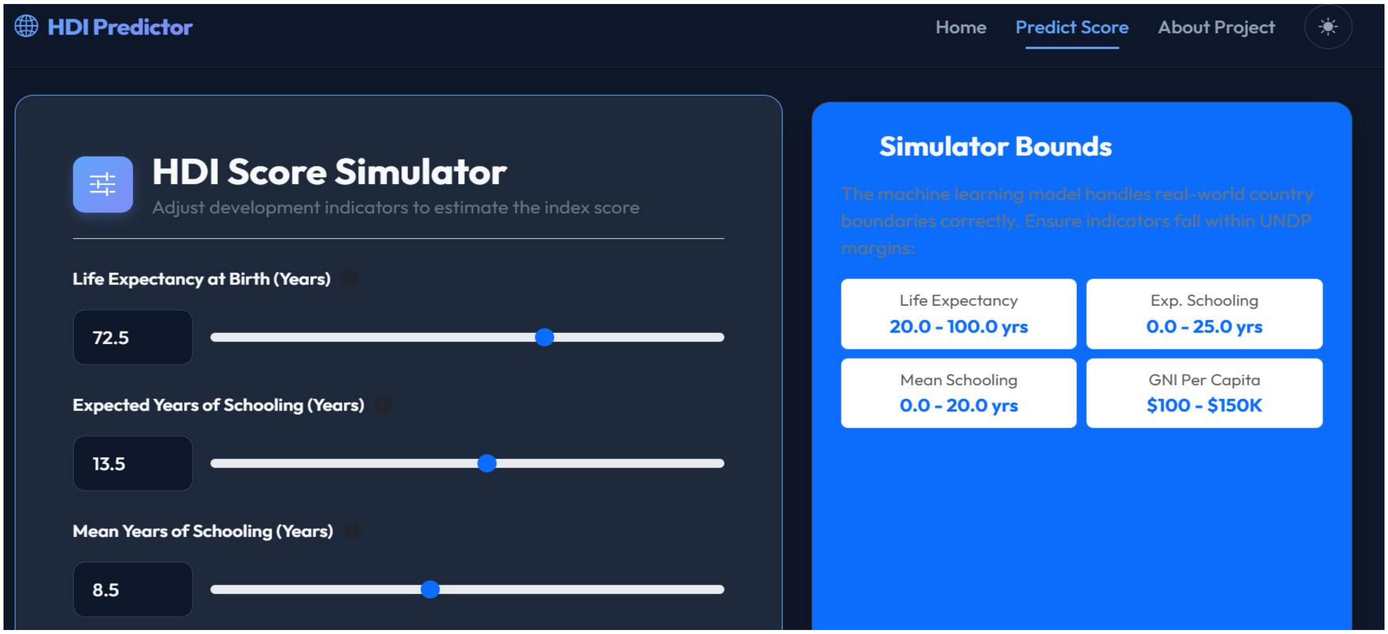



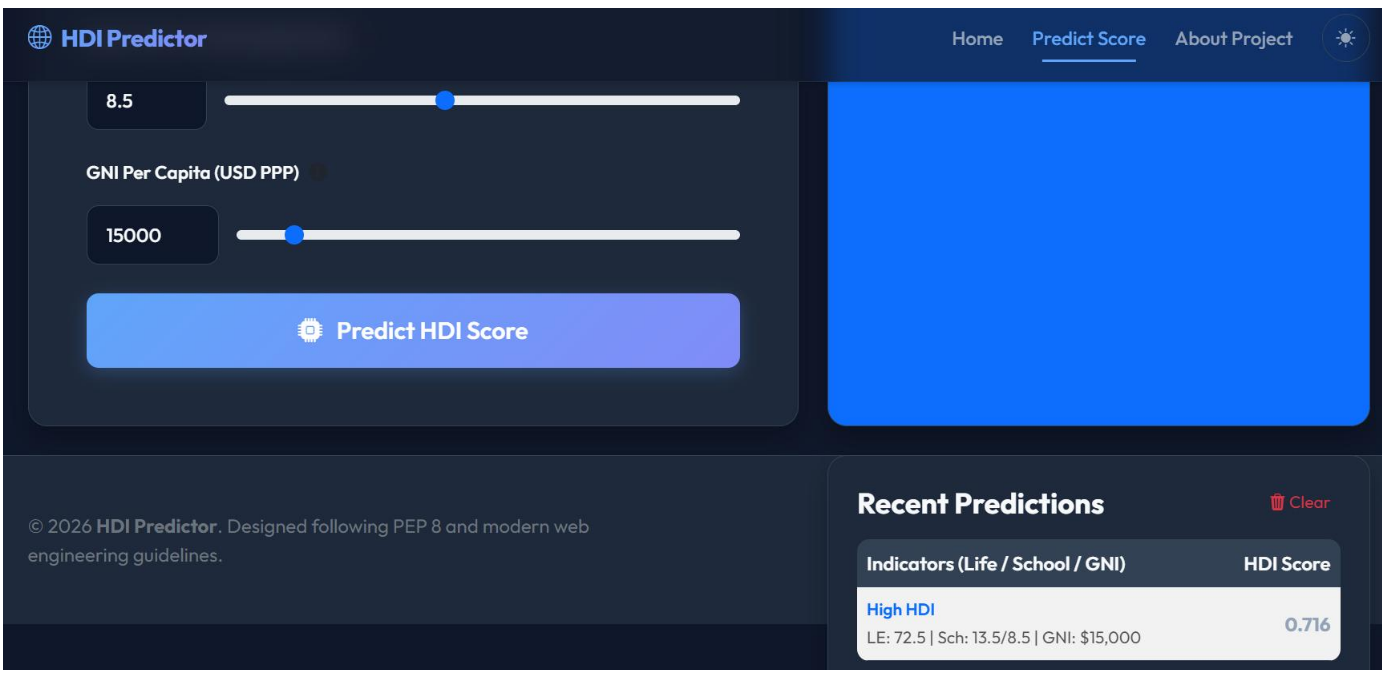



## Home Page Description:

The Home page serves as the landing page of the Human Development Index (HDI) Predictor, introducing the purpose and key features of the application. It provides an overview of the machine learning model along with sample prediction information to help users understand its functionality. Users can navigate to the prediction page to estimate the HDI score or learn more about the project through the About section.

## Input Section:

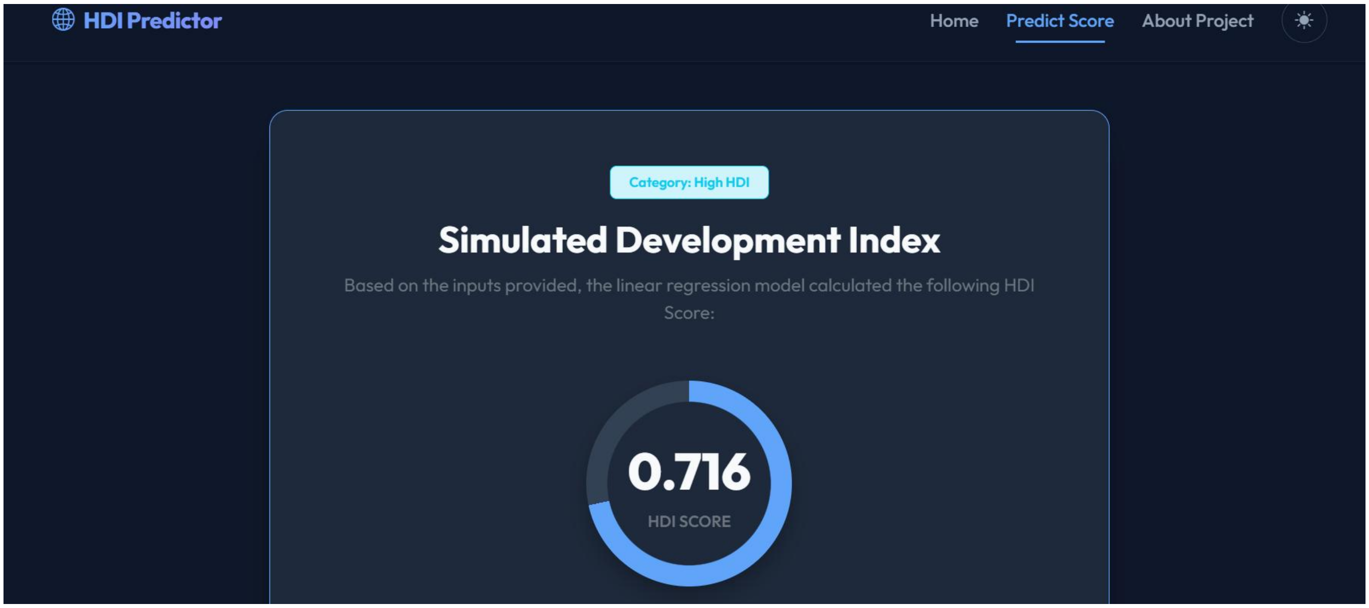


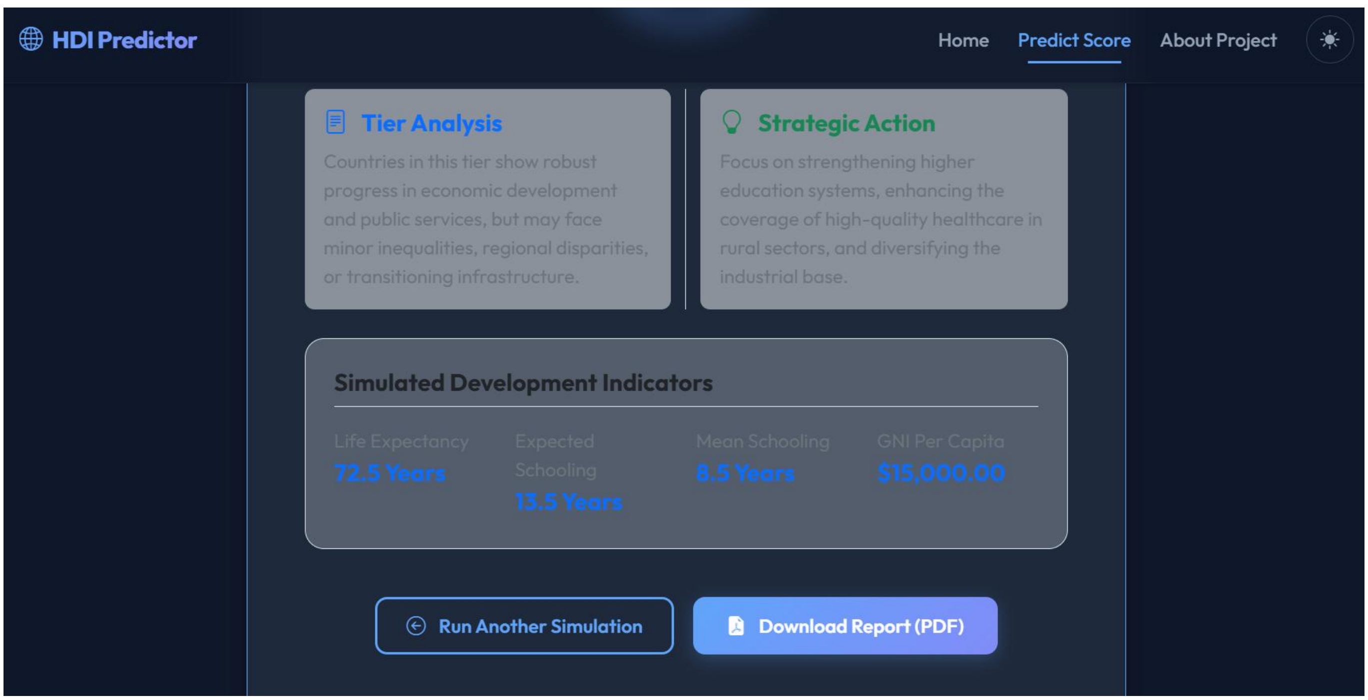




## Prediction Page Description

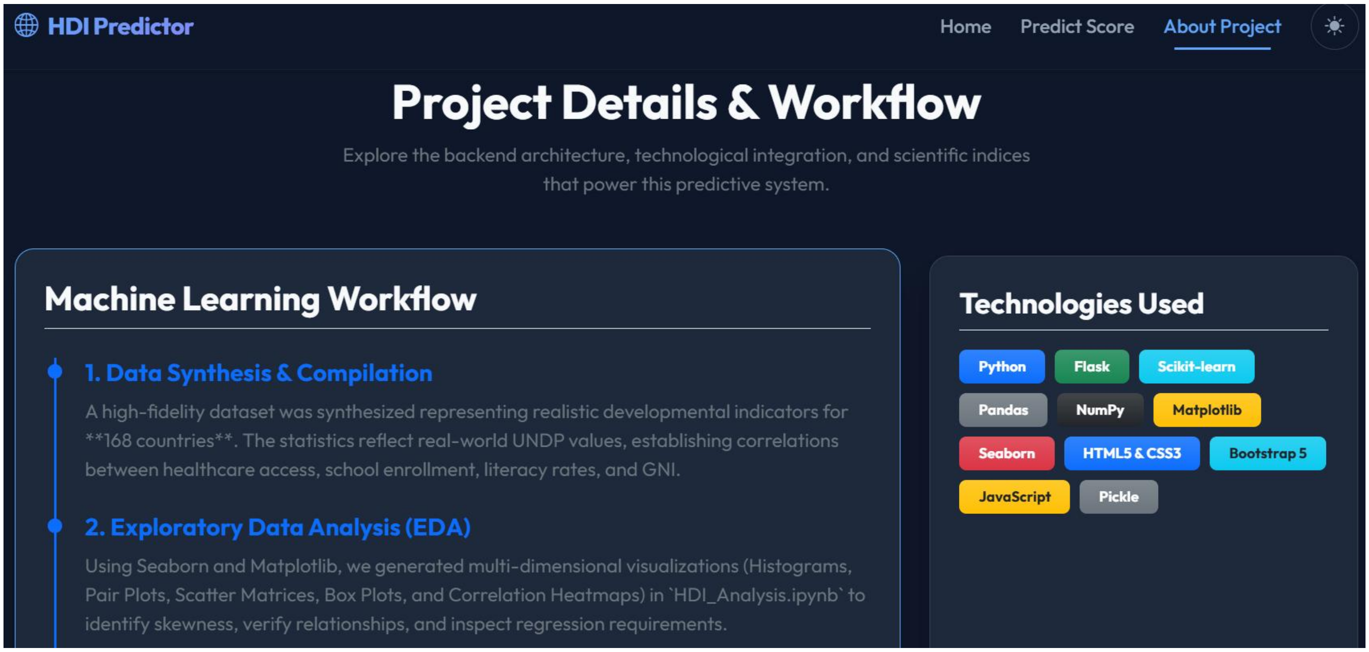
The Prediction page allows users to estimate the Human Development Index (HDI) score by entering key development indicators, including Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and GNI per Capita. The application validates the input values, processes them using the trained Linear Regression model, and generates an HDI prediction. The predicted score is displayed instantly along with the development category, while recent predictions are maintained for quick reference.

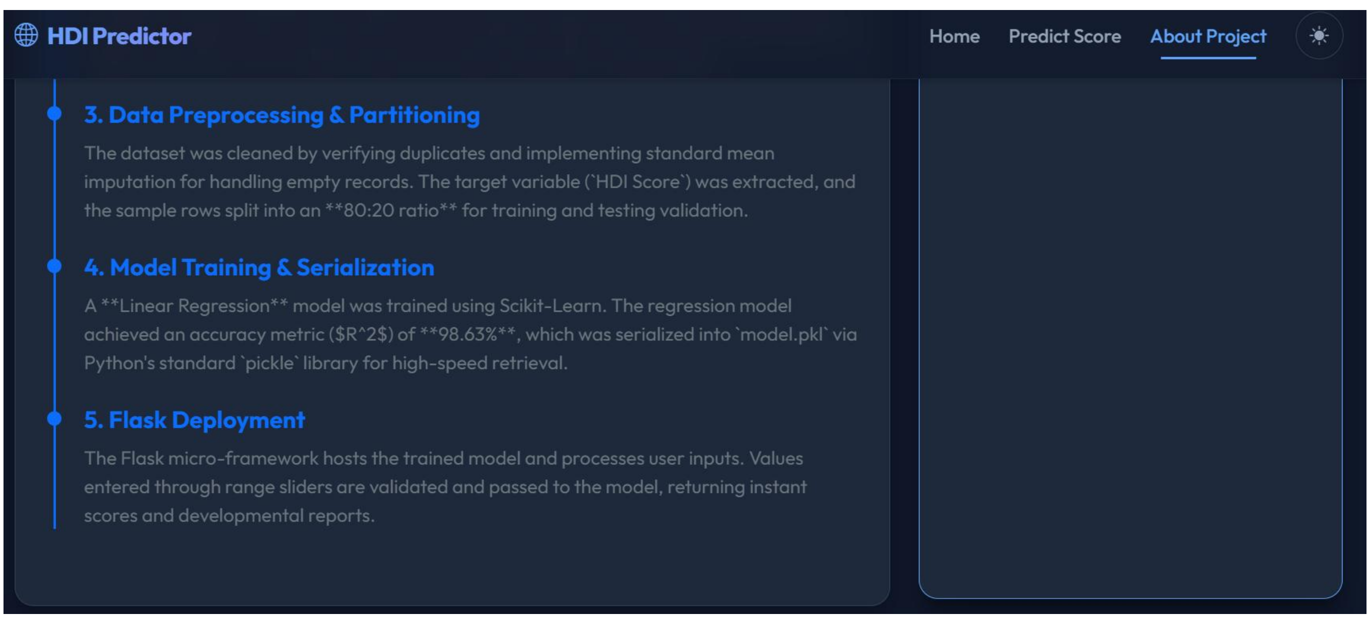
## Results Section:



## About Section :



# Conclusion :

The **Human Development Index (HDI) Predictor** successfully demonstrates the practical application of Machine Learning and web technologies in predicting a country's level of human development. By utilizing key socio-economic indicators such as **Life Expectancy**, **Expected Years of Schooling**, **Mean Years of Schooling**, and **Gross National Income (GNI) per Capita**, the application accurately estimates the HDI score using a **Linear Regression** model. The project integrates data preprocessing, exploratory data analysis, model training, evaluation, and deployment into a user-friendly Flask-based web application, enabling users to obtain predictions quickly and efficiently.

This project highlights how machine learning can support data-driven decision-making by providing valuable insights into the factors influencing human development. It can serve as a useful tool for students, researchers, policymakers, and development organizations to analyze development indicators and understand their impact on HDI. In the future, the application can be enhanced by incorporating additional socio-economic indicators, supporting multiple machine learning algorithms for improved prediction accuracy, integrating real-time datasets, and deploying the system on cloud platforms to make it more scalable, accessible, and beneficial for a wider range of users.